1번) 다음 그림은 DC 모터 1개와 조도 센서 1개를 사용하여 일정한 세기의 빛이 들어오면 빛의 세기를 DC 모터의 속도로 mapping하여 제어하는 tinkercad 회로의 예시이다. 각각의 조도센서와 DC 모터가 서로 연동이 되도록 회로 및 코드를 작성하라. (180점)

그림에서 조도센서에 연결된 저항값은 10 kΩ 임, RGB LED 연결저항값 220옴

```
const int EN=11;    //Half Bridge 1 Enable
const int MC1=2;  //Motor Control 1
const int MC2=3;  //Motor Control 2
const int LIGHT_SENSER=1;  //POT on Analog Pin 0
const int MIN_LIGHT = 200;
const int MAX_LIGHT = 1000;
const int RED    = 6;
const int GREEN = 4;
const int BLUE  = 5;
const int SPEED_MIN = 100;
const int SPEED_MAX = 200;

int val = 0;        //for storing the reading from the POT
int velocity = 0; //For storing the desired velocity (from 0-255)
int light = 0;

void setup()
{
    pinMode(EN, OUTPUT);
    pinMode(MC1, OUTPUT);
    pinMode(MC2, OUTPUT);
    brake(); //Initialize with motor stopped
    Serial.begin(9600);
    pinMode(RED, OUTPUT);
    pinMode(GREEN, OUTPUT);
    pinMode(BLUE, OUTPUT);

  //Turn off the LED
  //It is common-anode, so setting the cathode pins to HIGH turns the LED off
    digitalWrite(RED, LOW);
    digitalWrite(GREEN, LOW);
    digitalWrite(BLUE, LOW);
}

void loop()
{
    val = analogRead(LIGHT_SENSER);
    delay(50);
```

```
    Serial.print("₩nlight: ");
        Serial.print(val);

    //go forward
    if (val >= 500 && val <= 870)
    {
        velocity = map(val, 500, 870, 100, 200);
        forward(velocity);
        Serial.print("₩nforward: ");
        Serial.print(velocity);
        light = map(velocity, 100, 200,0,255);
        analogWrite(RED, light);
        digitalWrite(GREEN, LOW);
        digitalWrite(BLUE, LOW);
    }

    //go backward
    else if (val >= 930 && val <= 960)
    {
        velocity = map(val, 930, 960, 100, 200);
        reverse(velocity);
        Serial.print("₩nbackward: ");
        Serial.print(velocity);
        light = map(velocity, 100, 200,0,255);
        digitalWrite(RED, LOW);
        digitalWrite(GREEN, LOW);
        analogWrite(BLUE, light);
    }

    //brake
    else
    {
        brake();
        digitalWrite(RED, LOW);
        digitalWrite(GREEN, HIGH);
        digitalWrite(BLUE, LOW);
    }
}

//Motor goes forward at given rate (from 0-255)
void forward (int rate)
{
```

```
    digitalWrite(EN, LOW);
    digitalWrite(MC1, HIGH);
    digitalWrite(MC2, LOW);
    analogWrite(EN, rate);
}

//Motor goes backward at given rate (from 0-255)
void reverse (int rate)
{
    digitalWrite(EN, LOW);
    digitalWrite(MC1, LOW);
    digitalWrite(MC2, HIGH);
    analogWrite(EN, rate);
}

//Stops motor
void brake ()
{
    digitalWrite(EN, LOW);
    digitalWrite(MC1, LOW);
    digitalWrite(MC2, LOW);
    digitalWrite(EN, HIGH);
}
```